

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology	
Subject: Digital Signal and Image Processing(01CT1513)	To perform gray level operations on an image using Python.	
Exp-7	Date:	Enrollment No:92400133138

Aim:- To perform gray level operations on an image using Python.

Code :-

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

[1] ✓ 9.6s

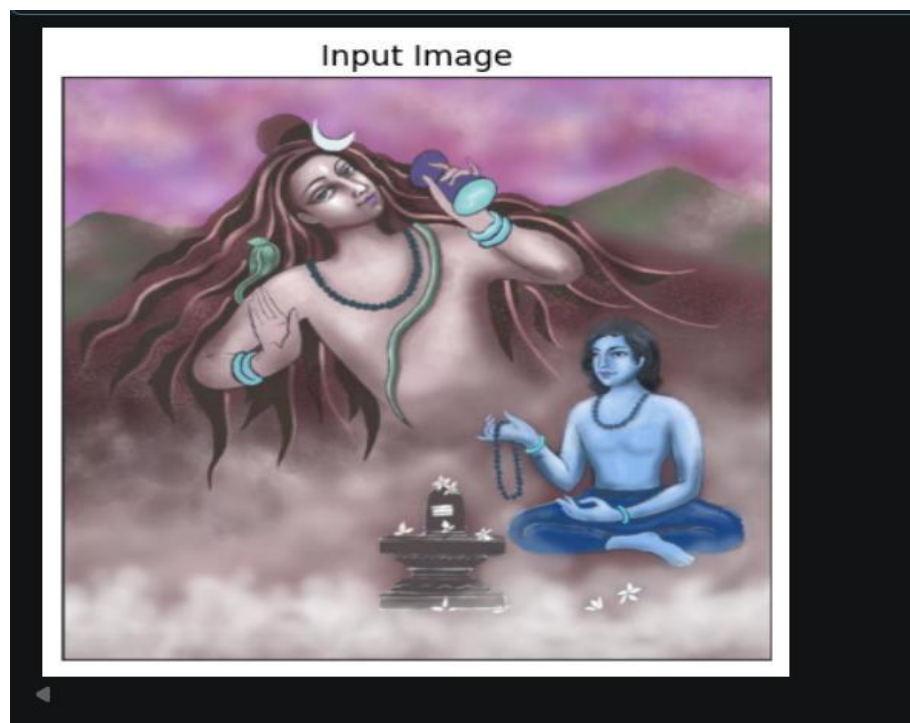
image = cv2.imread(r"C:\Users\dhruv\Pictures\Screenshots\Screenshot 2024-05-21 121309.png")

[2] ✓ 0.0s

plt.figure(figsize=(6, 6))
plt.imshow(image)
plt.title("Input Image")
plt.axis('off')
plt.show()

[3] ✓ 0.7s
```

OUTPUT:-



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```

img2 = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
✓ 0.0s
Generate Code

plt.figure(figsize=(6, 6))
plt.imshow(img2)
plt.title("Input Image")
plt.axis('off')
plt.show()
✓ 0.2s

```

Input Image



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Code :-

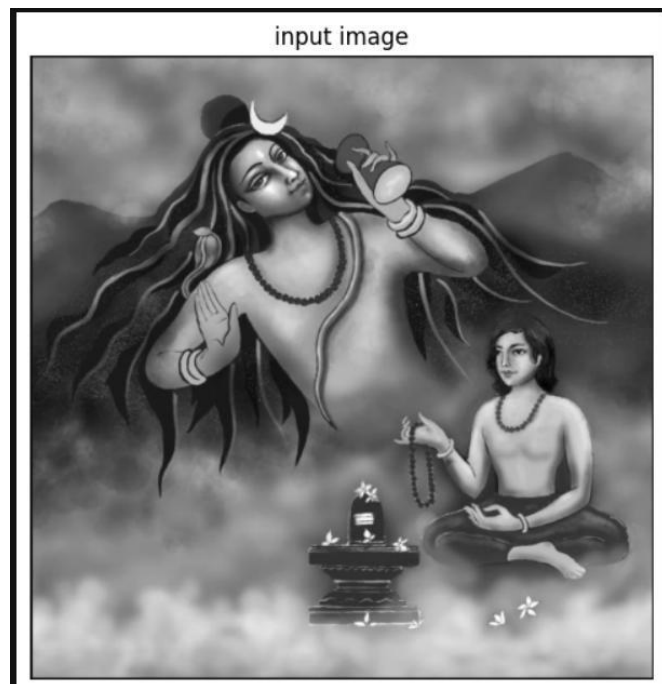
```

gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
✓ 0.0s

plt.figure(figsize=(6, 6))
plt.imshow(gray_image, cmap="gray")
plt.title("input image")
plt.axis("off")
plt.show()
✓ 0.3s

```

Output :-



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Code :-

```
c = 255 / np.log(1+ np.max(gray_image))
log_image = c* np.log(1 + gray_image)
```

```
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

✓ 0.0s

```
log_image = c * np.log(1 + gray_image)
```

✓ 0.0s

```
negative_image = 255 - gray_image
```

✓ 0.0s

```
plt.figure(figsize=(6, 6))
plt.imshow(negative_image, cmap="gray")
plt.title("Negative Image")
plt.axis("off")
plt.show()
```

✓ 0.3s

Output :-

Negative Image



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Code :-

```
_, thresholded_image = cv2.threshold(
    gray_image,
    128,
    255,
    cv2.THRESH_BINARY
)
✓ 0.0s

plt.figure(figsize=(6, 6))
plt.imshow(thresholded_image, cmap="gray")
plt.title("Threshold Image")
plt.axis("off")
plt.show()
```

Output :-



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Code :-

```

gamma = 2.0

normalized_image = gray_image / 255.0

gamma_corrected_image = np.power(
    normalized_image,
    1 / gamma
)

gamma_corrected_image = np.uint8(
    gamma_corrected_image * 255
)
✓ 0.0s

plt.figure(figsize=(6, 6))
plt.imshow(gamma_corrected_image, cmap="gray")
plt.title("Gamma Corrected Image")
plt.axis("off")
plt.show()

```

Output :-

Gamma Corrected Image



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Conclusion:

Thus, the gray level operations were successfully performed on the image using Python, including grayscale conversion, image negation, thresholding, and gamma correction.